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[1]: import nltk

[3]: nltk.download('omw-1.4')

[nltk_data] Downloading package omw-1.4 to C:\Users\Nemtullah
[nltk_data]   Khan\AppData\Roaming\nltk_data...
[nltk_data]   Package omw-1.4 is already up-to-date!

[3]: True

[4]: nltk.download('wordnet')

[nltk_data] Downloading package wordnet to C:\Users\Nemtullah
[nltk_data]   Khan\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!

[4]: True

[6]: import re
import pandas as pd
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer, WordNetLemmatizer

[9]: documents = [

    'Natural language processing is transforming Artificial Intelligence.',
    'Machine learning model require clean and normalized text data.',
    'Tokenization, stemming, and lemmatization are impotant NLP preprocessing step.'

]

[11]: stop_words = set(stopwords.words('english'))
stemmer = PorterStemmer()
lemmatizer = WordNetLemmatizer()

[13]: def preprocess_text(text):

    # Normalization
    text = text.lower()
    text = re.sub(r'^a-z\s|$', '', text)

    # Tokenization
    tokens = word_tokenize(text)

    # Stop-word removal
    filtered_tokens = [word for word in tokens if word not in stop_words]

    # Stemming
    stemmed_tokens = [stemmer.stem(word) for word in filtered_tokens]

    # Lemmatization
    lemmatized_tokens = [lemmatizer.lemmatize(word) for word in filtered_tokens]

    return {
        'Original_Text': text,
        'Tokens': tokens,
        'After_Stopword_Removal': filtered_tokens,
        'Stemmed': stemmed_tokens,
        'Lemmatized': lemmatized_tokens
    }

[18]: result = [preprocess_text(doc) for doc in document]

[22]: df = pd.DataFrame(result)
df
```

	Original_Text	Tokens	After_Stopword_Removal	Stemmed	Lemmatized
0	natural language processing is transforming ar...	[natural, language, processing, is, transformi...	[natural, language, processing, transforming, ...	[natur, languag, process, transform, artifici...	[natural, language, processing, transforming, ...
1	machine learing model require clean and normal...	[machine, learing, model, require, clean, and,...	[machine, learing, model, require, clean, norm...	[machin, lear, model, requir, clean, normal, t...	[machine, learing, model, require, clean, norm...
2	tokenization stemming and lemmatization are im...	[tokenization, stemming, and, lemmatization, ar...	[tokenization, stemming, lemmatization, impotan...	[token, steme, lemmat, impot, nlp, preprocess,...	[tokenization, stemming, lemmatization, impotan...

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